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Methodology to classify the shape of reinforcement fillers: optimization, evaluation, comparison, and selection of models

Roberto Fernandez Martinez[1,*], Maider Iturrondobeitia[1], Julen Ibarretxe[1], and Teresa Guraya[1]

[1] *eMERG, College of Engineering of Bilbao, University of the Basque Country UPV/EHU, Alameda Urquijo s/n, 48013 Bilbao, Spain*

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ABSTRACT

This work applies statistical analysis, and classical and advanced machine learning algorithms to classify 7714 aggregates into four categories according to their shape. The aggregates under study are obtained from several grades of carbon black: Vulcan XC 605, Vulcan XC 72, CSX 691, Printex 25, N990, and N762. The classification of the shape is of great significance in order to explain and predict the end-use properties of the composite materials, like mechanical properties. The proposed approach combines transmission electron microscopy and automated image analysis to obtain the dataset of the morphological features that defines the shape of the aggregate, and statistical analysis and machine learning techniques to create the classification models using feature transformation and reduction, parameter tuning, and validation methods in order to achieve robust classification models. The best result is obtained from a classification tree based on evolutionary algorithms with a principal component analysis-based feature reduction that reports an acceptable accuracy, thereby validating both the final chosen model and the methodology.

Introduction

The final application of materials is strongly dependent on their properties. In the case of filler-reinforced composite materials, the morphological features of the filler determine many of the composite properties [1]. In the manufacturing of spherical fillers, single particles fuse with or collide against each other to form aggregates with different shapes [2], whose automatic classification is a complex problem due to the difficulties in characterizing the aggregates. The shapes of these aggregates are of great significance with regards to the properties of the composite materials and, consequently, reliable knowledge of the distribution of the aggregate shapes in pre-defined categories will help to predict the end-use properties of the materials. This is the case with the electrical or mechanical properties of rubber blends reinforced with carbon black.

Nowadays, carbon black suppliers offer a wide range of grades that differ from one another in many of their characteristics, one of them being the

Address correspondence to E-mail: roberto.fernandezm@ehu.es

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'structure.' The definition of this characteristic is ambiguous and is measured by indirect standardized tests. The structure provides average information on the morphology of these aggregates, their specific surface volume ratio and intra-unit occlusion capacity, both responsible for the reinforcement capacity of the filler [3]. A step forward in the enhancement of the characterization of the aggregate structure can be accomplished by direct observation and reliable measurement of their dimensions in transmission electron microscopy (TEM) images [4–9]. Some studies have proposed several morphological categories to classify the shape of the carbon black aggregates. The most popular among them is by Herd et al. [6]. This study classifies aggregates into four morphological categories: spheroidal, ellipsoidal, linear, and branched. Figure 1 shows an example of aggregates classified in these four morphological classes. The combination of direct measurement of the aggregate size and their distribution by class has been a target of research for a long time because it provides a more reliable assessment of the structure of the grades based on direct observation.

These works have reasonable statistical significance, but their application requires the manual processing of thousands of aggregates, which is both tedious and highly time consuming. Maas et al. [8] made the first attempt in predicting the category of a specific aggregate using discrimination analysis. In recent years, novel methods are being accomplished through the combination of digital analysis of TEM images and advanced computational techniques. Alongside other interesting results, it enables the detection of structural details of the aggregates that can be useful for their automatic classification into pre-defined morphological classes based on their shape [10–13].

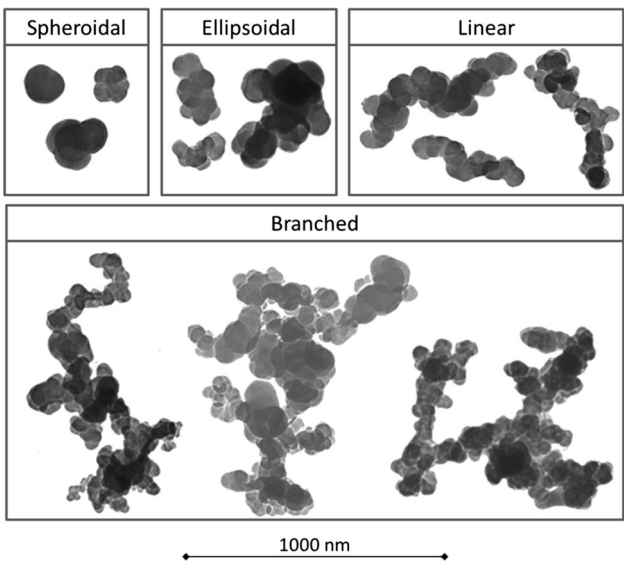

Figure 1 Morphological categories of carbon black aggregates.

Currently there are a large number of machine learning algorithms to classify a categorical response on the basis of a supervised dataset [14]. These classifying algorithms can be divided into three groups according to their operating methodology:

- Linear classification methods, which categorize samples into groups based on the predictor characteristics using intrinsically linear functions.
- Nonlinear classification methods, which seek to categorize samples in conditions where the class boundaries can be nonlinear functions.
- Classification trees and rule-based methods, which consist of if-then statements that split the feature space according to optimization criteria.

The main goal of this work is to build the most accurate model to automatically classify samples of aggregates extracted from any carbon black grade directly from TEM images, since the results obtained from the model will provide further information about the structure of the filler. To achieve this, several machine learning methods have been applied to a dataset, obtained automatically from the studied TEM images, that contains the geometrical and morphological features of a group of carbon black aggregates, selected to include a wide variation in size and structure. Several models are built and trained while feature reduction and transformation, and parameter tuning are performed. Later, the selected models are tested in order to identify their degree of generalization according to several robustness criteria [15–17]. And finally, samples of two other grades of carbon black, with different structures, are prepared to validate the models using aggregates that have not been used previously in the process.

Materials and methods

Materials

The carbon black grades presented in Table 1 were used to perform this work. Vulcan XC 605, Vulcan XC 72, and CSX 691 are high structure special grades

Table 1 Dataset used: number of TEM images, number of isolated aggregate images, and aggregate classification

Type of dataset	Grade of carbon black	Number of TEM images	Number of aggregates	Class of the aggregate			
				Spheroidal	Ellipsoidal	Linear	Branched
Train–test dataset	Vulcan XC 605, Vulcan XC 72, and CSX 691	62	2596	121	1240	446	789
	Printex 25	61	3117	232	2004	402	479
Validation dataset	N762-N990	19	2001	264	1244	116	377

Table 2 Features measured on the geometrical and morphological characterization [12]

Aggregate characteristics		
Defined in ASTM Standard D3849-07	Aggregate area	A
	Aggregate perimeter	P
	Diameter of a circle with the same area	D
	Aggregation factor	α
	Average particle size for an aggregate	d_p
	Aggregate volume	V_A
	Particle volume	V_p
	Number of particles in the aggregate	n
Based on the boundary	Roughness	Ro
	Signature	SSD
	Fourier descriptors	FD
	Ratio of number of segments	CCseg
	Total number of interceptions	CCcu
Based on the region	Circularity	C
	Solidity	So
	Extent	Ex
	Eccentricity	Ec
	Mean of extent	ExMean
	Standard deviation of extent	ExSD
	Mean of solidity	SoMean
	Standard deviation of solidity	SoSD

(more than 30 % of the aggregates are branched). They were supplied by Cabot Corporation in the form of high-quality lab samples, which were easy to deagglomerate during preparation. Printex 25 is also a special grade (15 % branched), but in this case the material came from the daily production material stock of a rubber blends manufacturer. N990 and N762 are formed by larger particles. They have lower structure (75 % (spheroidal + ellipsoidal)) and were also supplied by rubber manufacturers.

Morphological characterization and multivariate analysis

Material samples were prepared according to the ASTM standard D3849-07 [18], and TEM images were acquired and automatically processed following the methodology proposed by Fernandez Martinez et al. [12], in order to obtain twenty-one parameters that characterize the shape of each aggregate (Table 2).

Previously to the use of classification methods, all of the variables contained in the dataset were normalized between 0 and 1 to balance the weight of all variables and improve the statistical analysis quality. Subsequently, an exploratory analysis and statistical analyses of the variance (parametric and nonparametric tests depending on the feature: ANOVA, Bartlett, Brown–Forsythe, and Fligner–Killeen) and the correlation were carried out to determine the significance of each feature with the class. Also, these analyses looked for significant evidences of

Table 3 Classification techniques and corresponding references

Category	Classification technique	References
Linear classification	Linear discriminant analysis (LDA)	[19, 20]
	Partial least squares discriminant analysis (PLS)	[21, 22]
	Nearest shrunken centroids (NSC)	[23, 24]
Nonlinear classification	Mixture discriminant analysis (MDA)	[25]
	Flexible discriminant analysis (FDA)	[26, 27]
	Feed-forward neural networks (ANN)	[28]
	K-nearest neighbors (KNN)	[28]
	Support vector machines (with linear kernel (SVM-L), with polynomial kernel (SVM-P), and with radial basic function kernel (SVM-RBF))	[29–31]
Classification trees and rule based	Basic classification trees (BCT)	[32]
	C 5.0 (C50)	[33]
	C 4.5 (C45)	[34]
	Bagged CART (BCART)	[35]
	Random forest (RF)	[36]
	Evolutionary learning of globally optimal classification trees (evTree)	[37]

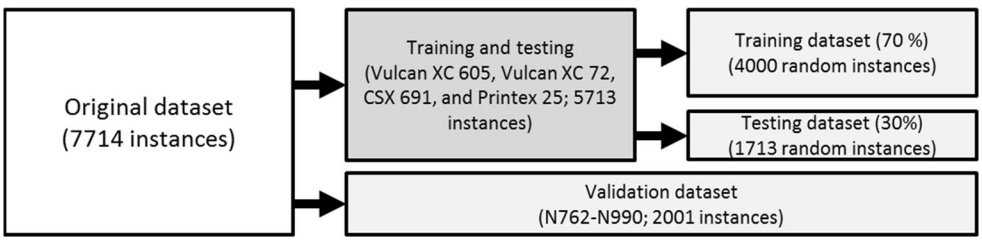

Figure 2 Methodology for the division of the dataset in order to create a successful process to train, test, and validate the models.

differences among the four classes and for possible redundant features.

The following steps of this methodology were applied twice, to the dataset with the original features and to the dataset with the new features obtained from the PCA analysis. PCA was applied to further reduce the dimension of the problem, since it is one of the most popular and successful methods for this proposal. The technique applies an orthogonal transformation of the initial space of variables, which creates a new set of variables that are named the principal components. The computed orthogonal projection captures the majority of the variance in the data and converts a set of possibly correlated variables into a new set of linearly uncorrelated components.

Classification methods

The proposed approach used several classification techniques to classify the aggregates by shape (Table 3). To perform the process, the dataset containing the selected features of the aggregates was used as input. Along the stages of building and training, feature reduction and parameter tuning were performed with the goal to simplify the models and achieve more robust classification models. The models obtained from this stage were tested with new data not used previously during the training in order to see the real generalization capacity of the models. With the results of this test, the models' performance was compared and a selection between the most accurate ones was conducted. And finally, a validation of the model with a new type of grade of carbon black was performed to see if the model really covered the entire range of possibilities of the problem avoiding any possible overtraining.

The whole process was performed according to the following methodology. The data were split into two groups according to the grades of carbon black: the training–testing dataset (5713 samples) and the validation dataset (2001 samples) (Fig. 2).

70 % of the samples from the training–testing dataset were randomly selected to train the models

using repeated ten-fold cross-validation. This training process was repeated 50 times since some methods, like ANNs, use random initializations (random initial values in their learning procedures). Therefore the process had to be repeated several times to improve the accuracy of the obtained results. During this training the most significant parameters of each algorithm were tuned to improve the results, and finally the most accurate models from each algorithm were selected for the testing stage.

At this point, the accuracy of the selected models was tested using an additional dataset (30 % of the samples previously split from the training–testing dataset), which had not been previously used for the training. This testing enabled the selection of the most accurate model in each case since their accuracy has been calculated with instances not used during the training stage.

After the most accurate models for each technique were chosen, they were validated using the validation dataset. The results of this validation showed that the chosen models successfully cover the whole spectrum of possibilities, and also showed the classification performance of the models with aggregates with really different structure. Based on these results, the most accurate model was selected as the best shape classifier of reinforcement fillers for composite materials.

Criteria of robustness

The robustness of the models is a commonly accepted parameter that helps to choose between them. It is mainly verified using the overall classification accuracy and its 95 % confidence interval as an objective function. It also takes other criteria into account when the overall classification accuracy is quite similar, i.e., unweighted kappa statistic, sensitivity, specificity, prevalence, detection rate, positive prediction value, negative prediction value, detection prevalence, and balanced accuracy (Table 4).

R statistical software environment v2.15.3 was used to program the proposed methodologies and to develop the classification models [38].

Results and discussion

Analysis of the features

The four proposed statistical tests to analyze the homogeneity of variance were performed for the 21 variables in the dataset, and both the statistics and the p-values were calculated for all of them. Additionally, a correlation analysis was performed as well to determine the dependence or association among features. In this case, it was concluded that the 21 features were significant for the aggregate image classification.

Therefore, they were introduced in a PCA and the results showed that it was possible to explain the 92.18 % of the variance of the data with just five Principal Components (PCs) (Table 5). The weight of the original features in each PC is reported in Table 6.

From this point, all the stages of the methodology were applied to both datasets, one with the original features and another with the new five selected PCs.

Table 4 Metrics used to optimize the classification model accuracy and select the best among them

$$\text{Accuracy} = \frac{TP+TN}{TI}$$
$$k = \frac{p_a+p_e}{1+p_e}$$
$$\text{Sensitivity} = \frac{TP}{TP+FN}$$
$$\text{Specificity} = \frac{TN}{FP+TN}$$
$$\text{Prevalence} = \frac{TP+FN}{TP+FP+FN+TN}$$
$$\text{PPV} = \frac{\text{Sensitivity}\cdot\text{Prevalence}}{(\text{Sensitivity}\cdot\text{Prevalence}+(1-\text{Specificity})\cdot(1-\text{Prevalence}))}$$
$$\text{NPV} = \frac{\text{Specificity}\cdot(1-\text{Prevalence})}{((1-\text{Sensitivity})\cdot\text{Prevalence}+\text{Specificity}\cdot(1-\text{Prevalence}))}$$
$$\text{Detection Rate} = \frac{TP}{TP+FP+FN+TN}$$
$$\text{Detection Prevalence} = \frac{TP+FP}{TP+FP+FN+TN}$$
$$\text{Balanced Accuracy} = \frac{\text{Sensitivity}+\text{Specificity}}{2}$$

TI total instances, *TP* aggregate has the condition and tests positive for the condition, *TN* aggregate does not have the condition and tests negative for the condition, *FP* aggregate does not have the condition but tests positive for the condition, *FN* aggregate has the condition but tests negative for the condition, p_a proportion of observations in agreement, p_e proportion in agreement due to chance

Results of the classification techniques

Following the proposed methodology, several models were built and trained using the training dataset. During this stage, the parameters that define the algorithm were tuned in order to determine the most suitable values to improve the accuracy. The techniques applied are listed in Table 7 together with the parameters that were tuned in the analysis, the range of tuning, and the selected values to build the final models. A ten-fold cross-validation process, which was repeated 50 times, was used to determine the classification accuracy of the models. The overall classification accuracy was the main criterion to analyze this accuracy but in case of doubt under similar error percentages the other explained criteria of robustness were studied as well.

Table 5 Importance of the first PCs (only the first five PCs selected are shown)

	PC1	PC2	PC3	PC4	PC5
Standard deviation	0.52	0.19	0.15	0.13	0.12
Proportion of variance	0.69	0.09	0.05	0.04	0.03
Cumulative proportion	0.69	0.78	0.83	0.88	0.92

The most accurate models for each technique from the training stage, in both cases: using the original variables and using the new PCs from PCA, were then tested with the testing dataset to analyze their real generalization capacity. For example, results obtained from the training and testing stages, when the dataset with all the features are used, are shown in Table 8.

The model based on ANN using all the original features obtained the most accurate results for aggregate classification during these stages (Table 9). However, it was noticeable that accuracy differences were not highly significant among the different techniques. Therefore it was decided to use the validation dataset without discarding any of the models at this point. Based on this, all the tested models were chosen to validate their real classification capability using the validation dataset.

Validation and selection of the most accurate model

To determine the most precise method and therefore, the most accurate model among all the options analyzed, a validation of the selected models was carried out. It is important to emphasize that the validation stage was carried out using the validation dataset

Table 6 Weight coefficients for each original variable of the new PCs

	Center	PC1	PC2	PC3	PC4	PC5
A	0.0275	−0.0641	−0.1575	−0.0559	0.0250	−0.0787
P	0.0381	−0.0771	−0.1416	−0.0388	0.0484	−0.0747
D	0.2650	−0.1708	−0.5317	−0.3741	−0.4171	0.0825
α	0.3547	0.4597	−0.0810	0.1262	0.0912	−0.3590
dp	0.1680	0.0391	−0.3148	−0.2039	−0.2651	−0.0121
Va	0.0209	−0.0557	−0.1640	−0.0641	0.0150	−0.0832
Vp	0.0159	0.0072	−0.0688	−0.0468	−0.0637	0.0016
N	0.0458	−0.1176	−0.1044	0.0302	0.1566	−0.0991
C	0.5126	0.3767	0.0070	0.0768	−0.0317	−0.1809
Ec	0.7263	−0.1273	0.4376	−0.7517	0.2798	−0.0553
Ex	0.5929	0.2471	0.0075	−0.2928	0.0167	−0.2023
So	0.7414	0.2784	−0.0187	−0.1031	−0.1039	−0.0195
SoMean	0.7835	0.2268	−0.0268	−0.0265	−0.0195	0.3714
SoSecSD	0.2703	−0.2461	0.2172	0.0969	−0.4305	−0.7000
ExMean	0.6089	0.2383	0.0188	−0.2705	−0.0229	−0.1753
ExSD	0.4832	−0.1938	0.4363	0.0648	−0.5023	0.2300
FD	0.0958	−0.0912	−0.1335	−0.0375	0.0426	−0.0550
CCseg	0.2137	−0.3518	−0.0746	0.1193	0.3201	−0.1150
CCcut	0.1104	−0.2059	−0.0782	0.0532	0.2394	−0.1269
Ro	0.1256	−0.1740	−0.1898	0.0369	0.1515	−0.0753
SSD	0.0926	−0.1511	−0.1668	−0.1258	0.0433	−0.0932

Table 7 Classification techniques applied in the analysis, with a brief definition of the parameters tuned during the training stage, and their selected values

Classification technique	Parameters	Grid of values	Selected values
LDA	None		
PLS	*Ncomp* the number of components to use for estimating the variances	1:21	17
NSC	*Threshold* values for the centroid shrinkage	1:30	3
MDA	*Subclasses* number of subclasses per class	1:8	1
FDA	*Degree* maximum degree of interaction (Friedman's mi)	1:5	2
	Nprune maximum number of terms (including intercept) in the pruned model	1:21	21
ANN	*Size* number of neurons in the hidden layer	1: 20	6
	Decay parameter to decrease the learning rate of the optimization function	0.001:0.1	0.1
KNN	*k* number of neighbors considered	1:20	17
SVM-L	*C* cost of constraints violation	0.2:10	6.2
SVM-P	*C* cost of constraints violation	0.2:10	0.35
	Degree degree of the polynomial kernel function	1:5	1
SVM-RBF	*C* cost of constraints violation	0.2:10	1.4
	sigma: inverse kernel width used by the Gaussian kernel	0.1:1	0.1
BCT	*Complexity parameter (CP)* any split that does not decrease the overall lack of fit by a factor of cp is not attempted	0:2	0
C50	*Trials* number of boosting iterations	1:100	50
	Model tree model or decomposed into a rule-based model	tree, rules	tree
	Winnow feature selector	yes, no	no
C45	*C* confidence threshold	0:1	0.21
BCART	None		
RF	*Mtry* number of variables randomly sampled as candidates at each split	1:21	1
evTree	*α (Complexity parameter)* any split that does not decrease the overall lack of fit by a factor of α is not attempted	0:3	1.5

that contains morphological features of carbon blacks extensively used during rubber blends manufacturing in the industrial processes. These carbon blacks are more difficult to prepare and characterize than the special grades used in the training and testing stages. But in this case, aggregates had a larger average size and exhibited a greater number of spheroidal and ellipsoidal shapes. These aggregate materials were selected for the validation process because their analysis gave a more reliable idea of the generalization capability of the models and the methodology.

The levels of accuracy obtained in the validation stage were a bit lower than the levels obtained during the testing stage, for example, Table 10 shows the results obtained when the dataset with all the features are used. These results can be compared with results from training and testing stages showed in Table 8. This is attributed to the significant differences between carbon black grades used for training and testing the models, and those used for validation. Even though the accuracy levels are lower, validating these models with data different from the data used to build them provides more meaningful results regarding the classification capacity, which covers the entire spectrum of possibilities of the aggregate shapes.

Based on all these results, the model finally selected was the one built by the method that combines variance and correlation analyses, PCA, and evolutionary learning of a globally optimal classification trees. The sequence of the methodology combined with the obtained results for this method is shown in the following section.

Results of the selected model

This section explains a more detailed sequence of the methodology, but in this case it is with the final selected model. Initially, the dataset containing 21 features for each of the 7714 aggregates was statistically analyzed, by means of variance and correlation analyses, to reduce the dimensionality of the problem. In this case, it was concluded that the 21 features

Table 8 Results obtained during the training and testing stages using all the feature dataset

Classification technique	Training		Testing		
	Accuracy (%)	Accuracy SD (%)	Accuracy (%)	95 % confidence interval	Kappa
LDA	**74.40**	**1.90**	**75.10**	**(72.96, 77.17)**	**0.56**
PLS	72.00	1.36	72.65	(70.44, 74.77)	0.54
NSC	72.10	1.73	71.75	(69.52, 73.90)	0.50
MDA	74.40	1.99	75.10	(72.96, 77.17)	0.56
FDA	74.70	1.99	74.81	(72.65, 76.87)	0.56
ANN	**74.80**	**1.87**	**75.70**	**(73.57, 77.75)**	**0.57**
KNN	73.30	1.85	74.57	(72.40, 76.64)	0.54
SVM-L	74.90	1.60	75.52	(73.39, 77.57)	0.55
SVM-P	74.90	1.67	75.46	(73.33, 77.51)	0.55
SVM-RBF	74.60	1.76	75.22	(73.08, 77.25)	0.55
BCT	69.80	2.12	66.77	(64.45, 69.03)	0.44
C50	**74.30**	**1.84**	**74.57**	**(72.40, 76.64)**	**0.56**
C45	73.80	1.85	73.61	(71.42, 75.71)	0.53
BCART	73.30	1.87	73.55	(71.36, 75.65)	0.54
RF	74.60	1.95	74.39	(72.22, 76.47)	0.54
evTree	73.21	1.40	72.17	(69.95, 74.31)	0.49

Bold letters denote results for the most accurate model for each of the three groups of techniques

Table 9 Confusion matrix and statistics obtained from the model based on ANN using all the features during the testing stage

	Spherical	Ellipsoidal	Linear	Branched
Spherical	45	28	0	1
Ellipsoidal	49	835	94	63
Linear	1	36	115	23
Branched	0	67	43	267
Sensitivity	0.4736	0.8644	0.4563	0.7542
Specificity	0.9815	0.7061	0.9576	0.9162
PPV	0.6081	0.8021	0.6571	0.7082
NPV	0.9686	0.7907	0.9081	0.9326
Prevalence	0.0569	0.5795	0.1511	0.2124
Detection rate	0.0269	0.5009	0.0689	0.1602
Detection prevalence	0.0443	0.6245	0.1049	0.2262
Balanced accuracy	0.7276	0.7853	0.7069	0.8352

were significant for aggregate classification. Therefore, they were introduced in a PCA and the results showed that it was possible to explain the 92.18 % of the variance of the data with just five Principal Components (PCs) (Table 5). Therefore, the first five PCs were used to build, train, and test, in this case a decision tree based on evolutionary algorithms (evtree).

After the dataset processing, the classification tree was implemented according to the proposed approach. During the training stage, the parameters to build the most accurate tree were tuned and at the same time the trees were optimized using genetic algorithms. Figure 3 shows the evolution of the accuracy when the complexity parameter (α) of the tree varies. Also, parameter like the minimum number of observations that must exist in a node in order for a split to be attempted (100), the minimum number of observations in any terminal node (200), the maximum depth of any node of the final tree (9), and the percentages of the population that is used in selection (20 %), pruning (20 %), major split rule mutation (20 %), minor split rule mutation (20 %), and crossover (20 %) are analyzed to optimize the results of each built tree.

The tree training process showed that the best performance was obtained when the tree was not pruned ($\alpha = 0$). The general training accuracy was 74.81 % with a 95 % confidence interval of (73.45, 76.15 %). However, several pruned trees reported similar results with the advantage of being easier to understand. The best performance obtained for a pruned tree ($\alpha = 1.5$) resulted in general training accuracy values of 73.21 % with a 95 % confidence interval of (71.81, 74.57 %). These values were similar to those obtained for the unpruned tree but with the advantage that the pruned tree used only two PCs.

The best pruned tree (Fig. 4) was subsequently selected and tested with the testing dataset. The

Table 10 Results obtained during the validation stage using all the feature dataset

Classification technique	Accuracy (%)	95 % confidence interval	Kappa
LDA	46.58	(44.37, 48.79)	0.13
PLS	47.93	(45.72, 50.14)	0.16
NSC	55.32	(53.11, 57.52)	0.22
MDA	46.58	(44.37, 48.79)	0.13
FDA	45.18	(43.08, 47.49)	0.14
ANN	52.72	(50.51, 54.93)	0.21
KNN	50.21	(48.00, 52.43)	0.18
SVM-L	53.42	(51.21, 55.63)	0.20
SVM-P	53.37	(51.16, 55.58)	0.19
SVM-RBF	59.97	(57.78, 62.13)	0.27
BCT	49.63	(47.41, 51.84)	0.16
C50	56.82	(54.62, 59.01)	0.26
C45	56.11	(53.90, 58.29)	0.21
BCART	50.22	(48.01, 52.44)	0.18
RF	53.62	(51.41, 55.83)	0.20
evTree	**63.77**	**(61.62, 65.88)**	**0.29**

Bold letters denote results for the most accurate model

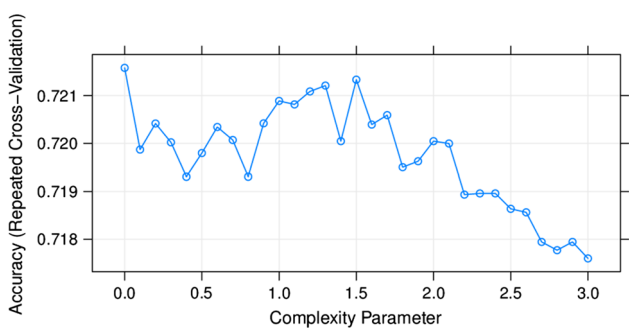

Figure 3 Variation of the accuracy (repeated cross-validation) on the selected classification model in relation to the complexity of the tree.

results obtained from this test are shown in Table 11. During the testing stage the general accuracy obtained was 72.17 % with a 95 % confidence interval of (69.95, 74.31 %) and a kappa value of 0.49. All these values were similar to those obtained during the training stage.

Finally and to obtain the real classification capability of the model, the model was validated. The overall accuracy obtained with this model was 73.21 % during the training stage, 72.17 % during the testing stage, and 63.77 % during the validation stage. Table 12 shows the confusion matrix and the statistics obtained using the evtree-based

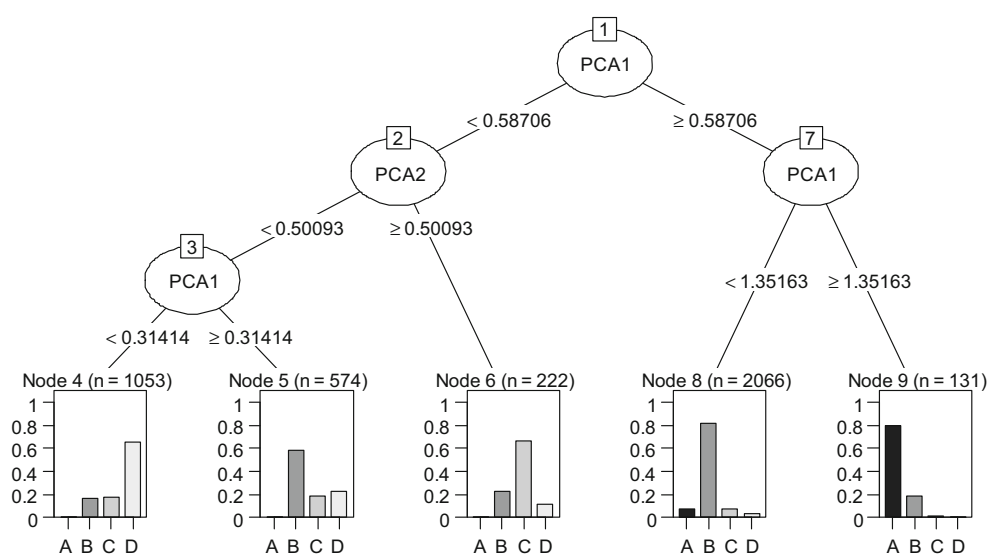

Figure 4 Generated classification tree that assigns carbon black aggregates to the four studied classes. *A* spherical, *B* ellipsoidal, *C* linear, and *D* branched.

Table 11 Confusion matrix and statistics of the selected technique (evtree) during the testing stage

	Spherical	Ellipsoidal	Linear	Branched
Spherical	38	20	1	1
Ellipsoidal	54	853	122	74
Linear	0	18	50	17
Branched	3	75	79	262
Sensitivity	0.4000	0.8830	0.1984	0.7401
Specificity	0.9860	0.6434	0.9752	0.8804
PPV	0.6333	0.7733	0.5882	0.6253
NPV	0.9645	0.7996	0.8723	0.9263
Prevalence	0.0569	0.5795	0.1511	0.2124
Detection rate	0.0228	0.5117	0.0299	0.1572
Detection prevalence	0.0359	0.6617	0.0509	0.2513
Balanced accuracy	0.6930	0.7632	0.5868	0.8103

Table 12 Confusion matrix and statistics of the selected technique (evtree) during the validation stage

	Spherical	Ellipsoidal	Linear	Branched
Spherical	198	244	0	0
Ellipsoidal	66	999	83	298
Linear	0	0	0	0
Branched	0	1	33	79
Sensitivity	0.7500	0.8031	0.0000	0.2095
Specificity	0.8595	0.4095	100.00	0.9790
PPV	0.4479	0.6909		0.6991
NPV	0.9576	0.5586	0.9420	0.8421
Prevalence	0.1319	0.6217	0.0579	0.1884
Detection rate	0.0989	0.4993	0.0000	0.0394
Detection prevalence	0.2208	0.7226	0.0000	0.0564
Balanced accuracy	0.8047	0.6063	0.5000	0.5943

methodology for analysis of the validation dataset. It is shown that spherical, ellipsoidal, and branched shapes are predicted with high values of accuracy, while with the linear shapes the produced error is quite high.

Due to the fact that the final classification tree only used two PCs, the results could be plotted in a two-dimensional graph (Fig. 5). In this graph, the tree splits the space of possibilities into five different regions according to class. It shows where the training, testing, and validation data are located in the space and how well the tree classifies each aggregate.

Conclusions

In this work, a three-step methodology that applies data mining processes based on data analysis and machine learning techniques has been validated. Its aim is to create models that can automatically classify any sample of carbon black according to its morphology based on TEM images. Firstly, a combination of transmission electron microscopy and automated image analysis was applied in order to obtain automatically a dataset that contained the geometrical and morphological features of 7714 aggregates. The dataset was split into two subsets. The first was used for the training and testing stages and consisted of 5713 special grade aggregates. The second was used for the validation stage and consisted of 2001 aggregates supplied from industrial users and which had a different class distribution. Secondly, a statistical analysis of the data, that included exploratory analysis, and variance and correlation analyses, was carried out in order to discard non-significant variables. In the third step, several classification models were built and optimized with feature reduction and parameter tuning techniques, to finally be validated in order to achieve robust classification models.

In order to assess performance of the methodology, the optimized models were validated, and the reported results show how larger and more heterogeneous datasets were better at defining the space of possibilities, since the decision boundaries of the classes were calculated more accurately than when using smaller datasets, where further improvements may be hindered by the lack of data.

Finally, once the most accurate model was found out and in light of the excellent results obtained, it was concluded that the methodology was successful in characterizing the carbon black filler and classifying into specific morphological categories. Also, the previous feature reduction resulted in a model that only needed two variables to classify the aggregates into the four classes. The model enabled the explanation of the problem in a two-dimension chart that was user friendly and easy-to-read. Thus, it was possible to determine the shape of these particles and consequently report useful information on how the filler affects the final properties of the composite material.

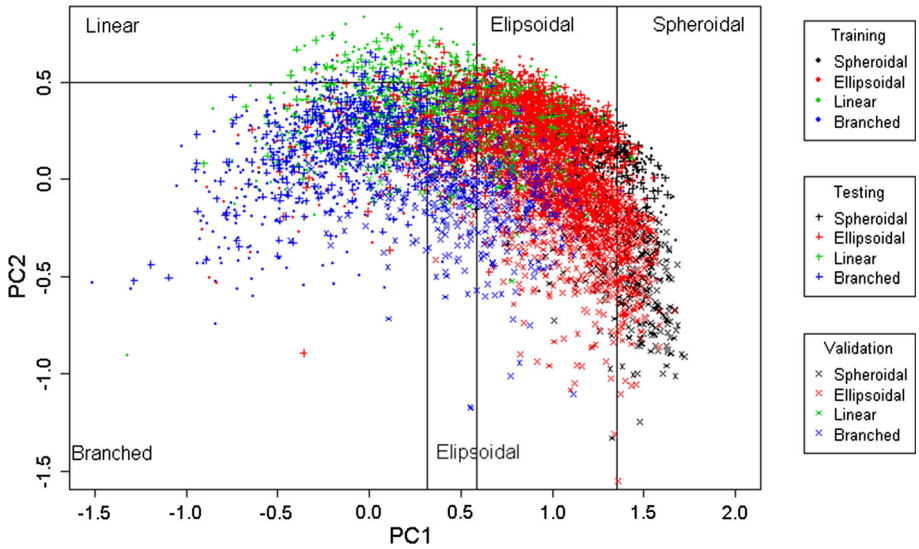

Figure 5 Classification of the aggregate dataset according to the selected model (evtree). Meaning of PC1 and PC2 according to Table 6.

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